

RESEARCH HIGHLIGHTS FOR FY1983*

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PRESENTED AT THE

NRC NOISE SURVEILLANCE AND DIAGNOSTICS
RESEARCH REVIEW GROUP MEETING
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RESEARCH HIGHLIGHTS FOR FY1983

- COMPLETED PWR PRESSURE NOISE ASSESSMENT INCLUDING MODEL OF PRESSURE NOISE IN PWR PRIMARY SYSTEM.
- COMPLETED SEQUOYAH DEMONSTRATION.
- INITIATED PEACH BOTTOM DEMONSTRATION.
- INITIATED AUTOMATED DIAGNOSTIC STUDY.
- ASSESSED LOOSE PARTS MONITORING.
- INITIATED STUDY OF TEMPERATURE NOISE.
- PROVED NEUTRON NOISE CAN BE USED TO ASSESS BWR STABILITY AND APPLIED TO DRESDEN.
- COMPLETED STUDY OF SENSITIVITY OF PWR EX-CORE NEUTRON NOISE TO FUEL ELEMENT AND ISSUED DRAFT NUREG REPORT.
- ISSUED DRAFT NUREG REPORT OF LWR NEUTRON NOISE.
- ASSESSED A PRESSURE PERTURBATION METHOD TO MEASURE RESPONSE TIME.

WHAT WE DID NOT ACCOMPLISH IN FY1983

- DID NOT STUDY ALL SEQUOYAH SIGNATURES IN DETAIL.
- DID NOT CONDUCT HIGH-PRESSURE TESTS OF PRESSURE SENSOR/SENSING LINES.
- DID NOT PURCHASE ARRAY PROCESSOR.
- DID NOT ISSUE LONG-RANGE RESEARCH PLAN.
- DID NOT PUBLISH REPORTS ON ORIGINAL SCHEDULE.

TASKS PERFORMED IN ADDITION TO THE FY1983 PLAN

- IN-CORE NEUTRON AND TEMPERATURE NOISE MEASUREMENTS AT SEQUOYAH-1
- NOISE MEASUREMENTS AT DRESDEN 2